

Architectural Elements Augmentation from 2D Artworks for Spatial Experience of AR Exhibition

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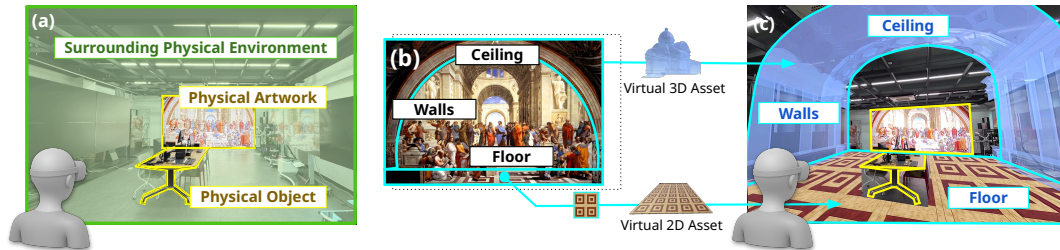


Figure 1: Conceptual and implementation visualization of the proposed system. (a) A user views a physical artwork in an indoor space. (b) Architectural elements (walls, ceiling, floor) are extracted from the artwork to generate virtual 2D/3D assets. (c) Real-time view showing seamless spatial augmentation without occluding physical objects.

ABSTRACT

We present a spatial augmentation system designed for Augmented Reality (AR) exhibition experiences, focusing on the immersive reinterpretation of two-dimensional artworks. Existing AR exhibition applications primarily augment content related to the artwork, but fall short of its architectural elements, which are crucial for integrating the artwork into the viewer's surrounding space and enabling immersive spatial alignment. This study proposes a spatial augmentation approach that takes an artwork image as input, generates floor patterns and architectural 3D models through 2D and 3D model generation pipelines, and semantically aligns and augments them in the physical space. This spatial augmentation intends to enhance viewer engagement and perceived presence by maintaining spatial coherence between virtual and real space. By activating architectural surfaces as part of the experience, the system repositions AR exhibitions as spatial narratives rather than conventional overlay-driven displays.

Index Terms: Augmented Reality, Spatial Coherence, Semantic Surface Alignment, AR exhibition.

1 INTRODUCTION

Augmented Reality (AR) has emerged as a compelling medium for exhibitions, allowing digital content to be superimposed within the physical space while preserving the user's spatial awareness. This capacity has led recent applications to augment physical exhibition spaces with architectural elements related to the artwork, such as annotations, symbolic figures, or narrative overlays, thereby enriching the interpretive experience without isolating users from the real-world, as Virtual Reality (VR) does [2]. Recent work has also addressed the engagement of cultural content through object-based

interaction and contextual overlays [7]. Despite this progress, most current systems do not extend the spatial context of the artwork, as they rarely augment architectural surfaces such as walls, ceiling, or floor due to the variability of exhibition spaces and the difficulty of standardizing spatial configurations for each user [4].

To address these limitations, we propose a spatial augmentation system that analyzes the spatial composition of 2D artworks and integrates their architectural elements into the surrounding physical space. Although existing systems often focus on object-level augmentation or floating overlays near artworks, such approaches overlook the spatial logic and continuity of the exhibition space. Previous research emphasizes that aligning virtual content with architectural context can enhance presence and perceptual coherence, especially in AR exhibition experiences where the physical world remains visible [3]. Wienrich et al. [6] further argue that spatial features serve as referential cues that shape spatial presence, suggesting that spatial augmentation is not merely decorative but perceptually essential. To fully realize the immersive potential of AR in exhibition contexts, it is essential to move beyond object-level augmentation and incorporate spatial augmentation that anchors these elements within the structural fabric of the physical space.

We demonstrate our system using Raphael's *The School of Athens*, which features rich architectural elements such as walls, arches, columns, domes, and a patterned floor that spatially align with real-world geometry and support coherent AR augmentation. Our system contributes by aligning virtual content with the architectural context of the exhibition space to enhance spatial coherence and ensure spatial continuity (see Figure 1). By using walls, floor, and ceiling as spatial anchors, it supports intuitive user orientation and strengthens spatial presence. Furthermore, architectural elements derived from the artwork are embedded into the surrounding space to preserve narrative flow and reduce perceptual separation between the virtual and physical space.

2 SYSTEM OVERVIEW

The system extends the spatial semantics of 2D artworks by reconstructing key architectural elements as 3D virtual assets and aligning them with real-world surfaces (see Figure 2). These elements are mapped onto walls, floor, and ceiling to integrate with the user's space. Our system was developed in Unity 6000.0.3f1 and deployed on Meta Quest 3 for the detection of semantic surfaces and precise

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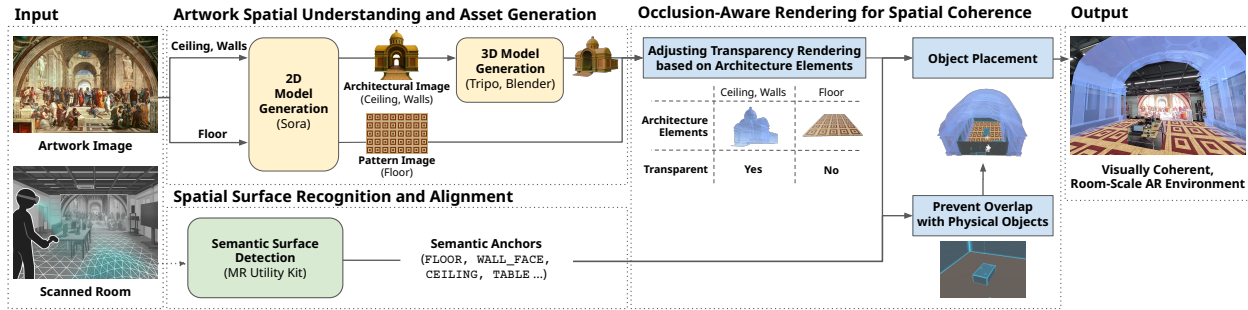


Figure 2: The system generates 2D and 3D architectural assets from artwork images and aligns them with structural surfaces using semantic anchors provided by the MR Utility Kit. These assets are rendered in Unity using surface-specific rendering to maintain spatial coherence.

spatial placement, utilizing the Mixed Reality Utility Kit (MRUK)¹.

Artwork Spatial Understanding and Asset Generation

In order to achieve spatial alignment between a 2D artwork and the physical space, it is essential to analyze the artwork's architectural elements and generate 3D assets that reconstruct its spatial structure. For the floor, tile patterns from the lower region of the artwork were extracted using OpenAI's Sora² and applied as textures on planar surfaces within Unity. Architectural components such as columns and a domed ceiling were generated through a two-stage pipeline involving 2D rendering and 3D reconstruction. These elements were first synthesized as frontal 2D images and then converted into 3D assets using Tripo³, an AI-based 2D-to-3D reconstruction tool. The resulting FBX files were refined in Blender for structural consistency and imported into Unity to align with the original painting's spatial composition.

Spatial Surface Recognition and Alignment

To ensure contextual alignment between virtual content and the physical space, the system preserves spatial understanding data captured by the Meta Quest 3's room scanning. This spatial data is transmitted to Unity via the MRUK, which provides semantic anchors labeled as FLOOR, WALL_FACE, CEILING, etc. These predefined categories allow the system to recognize planar surfaces and serve as reference points for placing virtual assets. By anchoring content to these surfaces, the system maintains spatial consistency and ensures that augmentation aligns with the user's space.

Occlusion-Aware Rendering for Spatial Coherence

The system dynamically assigns either transparent or opaque rendering to each architectural element. Fully opaque walls or ceiling can hinder spatial awareness, whereas alpha blending minimizes perceptual discontinuities and enhances MR coherence [5]. Walls and ceiling are rendered with transparency to maintain spatial continuity between the physical and virtual space. In contrast, the floor remains opaque to preserve a natural perception of depth and provide a grounded reference plane for the user. To prevent visual overlap between virtual content and physical objects, the system uses Unity's Spatial Mapping Occlusion, which gives visual priority to real-world geometry to maintain perceptual coherence [1].

3 CONCLUSION AND FUTURE WORK

This study presents a method to generate immersive AR experiences by extending the spatial composition of 2D artworks into physical space. By aligning architectural elements with real-world features such as walls, floor, and ceiling, the proposed system addresses the perceptual disconnect commonly found in existing AR

exhibition applications. Rather than simply overlaying content, this approach extends AR's role to spatial augmentation, allowing artworks to redefine and inhabit the physical space. By promoting spatial coherence between virtual and physical architectural elements, the system supports a more immersive, situated experience.

These findings open up possibilities for AR-based exhibition design, positioning architectural space as an active curatorial medium. Although shown with a single 2D artwork, the spatial augmentation strategy is broadly applicable across diverse exhibition formats and content types. Future work will extend the system to support a wider range of 2D artworks and adapt to diverse exhibition settings. Automated spatial enhancement features will be developed to reduce manual interventions. Additionally, empirical studies will assess the system's effects on immersion, spatial presence, and interpretive engagement.

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¹<https://developers.meta.com/horizon/documentation/unity/unity-mr-utility-kit-overview>

²<https://openai.com/sora>

³<https://studio.tripo3d.ai>